NETHOLABS

Data Quality Report

Functional Brain Emulation — 67 GB across 42 XDF recordings

March 2026

3	3	2
Critical Issues	Warnings	Info
Block analysis or reduce validity	May affect data quality	Cleanup and organization

ISSUES

CRITICAL Single Subject Dominance

GENERALIZABILITY

All 8 largest session recordings appear to be from the same participant. The 42 XDF files list 9 subject IDs, but the richest multi-modal data comes from undifferentiated `session_*` timestamped files with no subject identifier. Between-subject variability cannot be assessed with current data.

```
Subject distribution:
session_* (unnamed) — 14 files, 3.1 GB, 8 modalities each
sub-P001 — 6 files, 13.8 GB (includes 5 _old versions)
sub-P002 — 2 files, 3.7 GB (includes 1 _old)
sub-P007, P009, 090, 0903, 09031, 0999, 2003 — 1-3 files each, smaller
```

CRITICAL No Session Contains All Modalities

CROSS-MODAL ANALYSIS

The lab stack (fNIRS, GP3, StretchSense) and mobile stack (EEG, Tobii, Pupil Neon) have never been recorded simultaneously. No single XDF file contains all 11 modalities. Cross-stack correlations (e.g., EEG vs. fNIRS) cannot be computed from any existing recording. EMG (Delsys) has not appeared in any exported session.

```
Lab sessions contain: Gazepoint + KernelFlow (×3) + StretchSense + Microphone
Mobile sessions contain: DSI24 + TobiiGlasses (×2) + StretchSense
EEG never co-recorded with fNIRS
EMG (Delsys) absent from all exported sessions
```

CRITICAL

Kernel Flow Sample Rate Mismatch

SYNCHRONIZATION

All three KernelFlow LSL streams report a nominal rate of 5 Hz in their metadata, but pyxdf measured effective rates of 8–13 Hz across sessions. This 2–3× discrepancy causes LSL's timestamp correction algorithm to miscalculate, introducing synchronization drift relative to other streams.

```
Declared rate: 5.0000 Hz
Measured rates:
  KernelFlow_Red — 8.08 – 13.14 Hz
  KernelFlow_IR — 8.32 – 13.13 Hz
  KernelFlow_Quality — 0.99 – 13.62 Hz (one session dropped to ~1 Hz)
```

WARNING

Short Usable Overlap Windows

USABLE DURATION

Streams start and stop at different times within each recording, reducing the window where all modalities overlap. Some sessions yield under 40 seconds of usable multi-stream data despite large file sizes. Total usable overlap across 8 exported sessions: ~877 seconds (14.6 minutes).

```
session_20260306_171345.xdf — 37s overlap (22.5 MB file)
session_20260227_145319.xdf — <5s usable (241 KB file)
Best sessions (Mar 5-6): 120s+ overlap
```

WARNING

Gazepoint Rate Anomaly

DATA INTEGRITY

In one session, the Gazepoint stream was measured at 147 Hz instead of the expected 60 Hz. This suggests duplicate samples, a Gazepoint Control misconfiguration, or two stream instances merged into one recording.

```
Expected: 60 Hz   |   Measured: 147.19 Hz (~2.5× expected)
Other sessions: normal ~60 Hz
```

WARNING

_old File Accumulation

STORAGE & CLARITY

13 XDF files carry `_old1` through `_old5` suffixes, totaling ~15 GB. These are LabRecorder overwrites or manual re-recordings that may contain partial, corrupted, or superseded data. They inflate the total dataset size and create ambiguity about which files are canonical.

```
sub-P001_..._eeg_old1.xdf — 4.6 GB
sub-P001_..._eeg_old2.xdf — 7.1 GB
```

```
sub-P001_..._eeg_old3/4/5.xdf — 1.3 GB + 421 MB + 176 MB
+ 8 more _old files across other subjects
```

INFO

Aborted / Minimal Recordings

CLEANUP

Several XDF files are under 1 MB, likely from aborted recording attempts or quick equipment tests. These contain insufficient data for analysis and should be archived separately.

```
sub-P009 — 56 KB | session_20260305_150657 — 44 KB | session_20260227_145319 — 241 KB
sub-P001 (current, non-old) — 301 KB | sub-090_ses-090 — 426 KB
```

INFO

Inconsistent BIDS Naming

ORGANIZATION

Subject and session identifiers use mixed conventions: zero-padded alphanumeric (sub-P001, ses-S001), bare numeric (sub-090, ses-1), and no subject ID at all (session_YYYYMMDD_HHMMSS). Task labels also vary (Default vs. Baseline). This complicates automated BIDS-compliant processing pipelines.

```
sub-P001_ses-S001_task-Default_run-001_eeg.xdf — structured, zero-padded
sub-090_ses-010_task-Baseline_run-001_.xdf — different pattern, trailing underscore
sub-2003_ses-1_task-Baseline_run-001_.xdf — no zero-padding
session_20260306_091530.xdf — no subject ID
```

MODALITY COVERAGE BY SESSION TYPE

SESSION TYPE	EEG	FNIRS OXY	FNIRS DEOXY	FNIRS QA	GAZE (LAB)	GAZE (MOBILE)	EMG	HAND / BODY	IMU	AUDIO
Lab sessions (session_*)	—	✓	✓	✓	✓	—	—	✓	—	✓
BIDS subjects (sub-P00X)	?	✓	✓	✓	✓	—	—	✓	—	✓
Mobile sessions	✓	—	—	—	—	✓	—	✓	✓	—
Target (full coverage)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

FILE SIZE DISTRIBUTION (TOP 15)

FILE	SIZE	RELATIVE	STATUS
sub-P001_old2	7.1 GB		_OLD
sub-P001_old1	4.6 GB		_OLD
sub-P002_old1	3.7 GB		_OLD
sub-P007_ses-S007	1.7 GB		GOOD
sub-090_ses-030	1.5 GB		GOOD
sub-P001_old3	1.3 GB		_OLD
session_0306_091530	967 MB		GOOD
sub-09031_ses-1	732 MB		GOOD
session_0305_204932	643 MB		GOOD
session_0302_170251	550 MB		GOOD
session_0305_210738	508 MB		GOOD
sub-P001_old4	421 MB		_OLD
session_0306_181143	218 MB		GOOD
sub-P009	56 KB		EMPTY
session_0305_150657	44 KB		EMPTY

RECOMMENDATIONS

#	PRIORITY	ACTION	ADDRESSES
1	HIGH	**Run a combined lab + mobile session.** Connect DSI-24, Kernel Flow, Gazepoint, StretchSense, and Delsys simultaneously into a single LabRecorder instance. This is the only way to enable cross-modal EEG-fNIRS analysis.	Issues #1, #2
2	HIGH	**Fix Kernel Flow stream metadata.** Update `nominal_srate` in `kernel_flow_lsl.py` to match actual output (~10-13 Hz), or configure Kortex to output at a stable declared rate. Verify with `check_streams.py`.	Issue #3
3	HIGH	**Record 3+ additional subjects** with the full multi-modal stack. Target 5+ minute continuous recordings per session to establish between-subject baselines and enable group-level analysis.	Issue #1
4	MEDIUM	**Standardize BIDS naming.** Use consistent zero-padded subject/session IDs. Rename `session_*` files into `sub-XXX` folder structure. Archive _old files after verification.	Issues #6, #8

#	PRIORITY	ACTION	ADDRESSES
5	LOW	**Add pre-recording validation to `start_all.bat`** that verifies all streams are active and at expected rates before LabRecorder starts. This prevents empty/aborted recordings and catches rate anomalies.	Issues #4, #5, #7